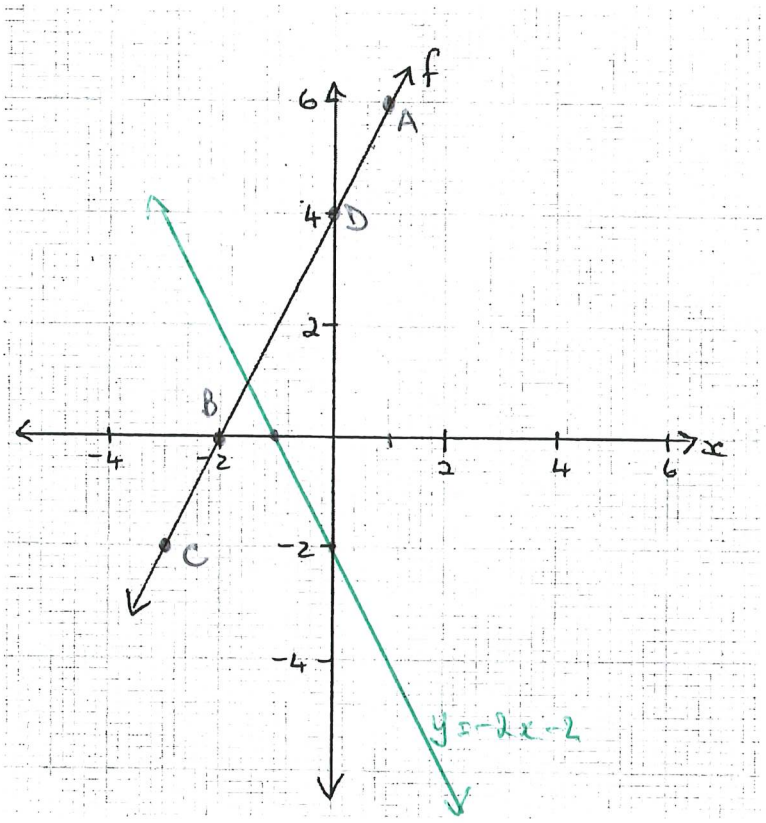
Graph Applications worksheet

EXERCISE A

1. On the set of axes below, the graph of $y = mx + c$ has been drawn.



1.1 Determine the equation of f: $y = 2x + 4$

1.2. **Use the graph** to answer the following questions:

Find the value of y if :

1.2.1 $x = 1$ $y = 6$ see A

1.2.2 $x = -2$ $y = 0$ see B

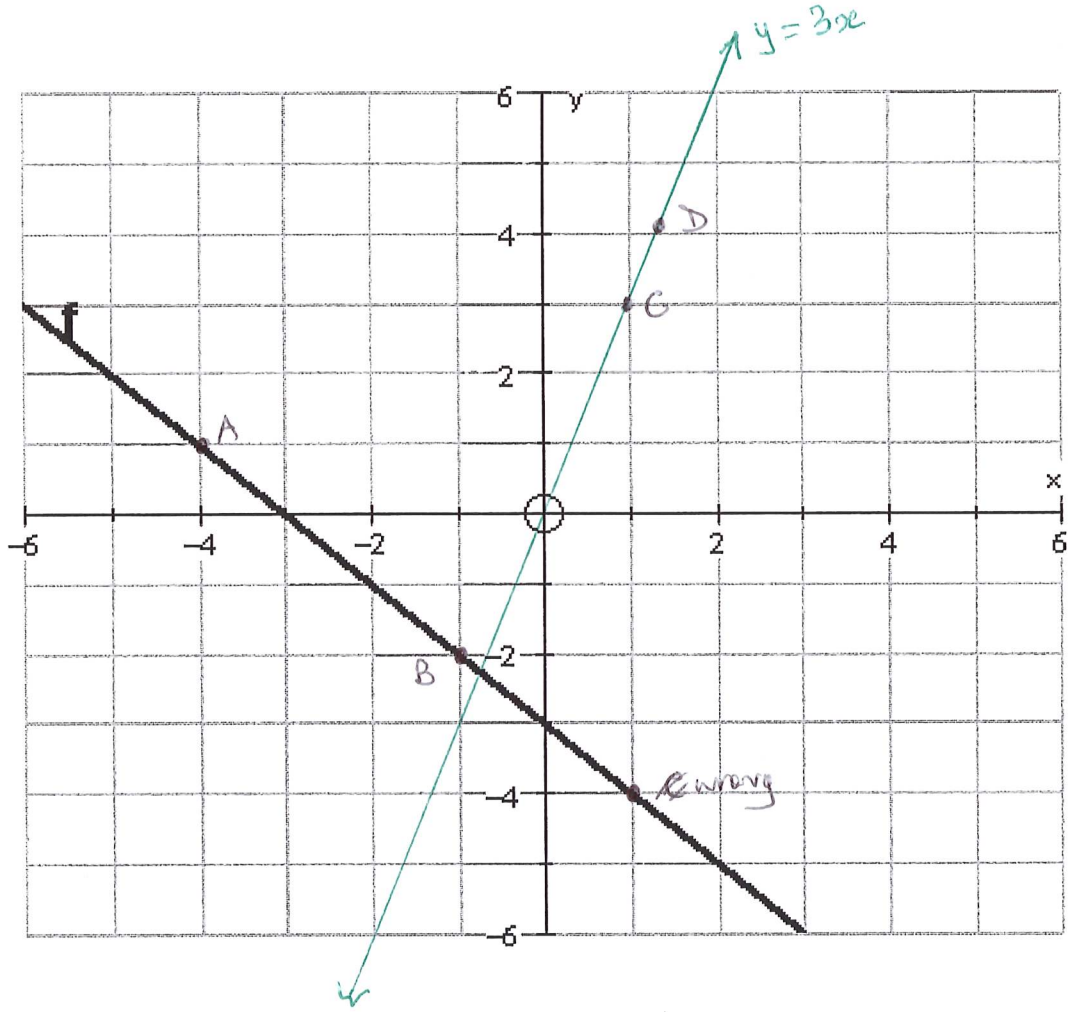
For what value(s) of x is:

1.2.3 $y = -2$ $x = -3$ see C

1.2.4 $y = 4$ $x = 0$ see D

1.3. On the same system of axes, draw the graph of $y = -2x - 2$

2. On the set of axes below, the graph of $y = mx + c$ has been drawn.



2.1 Determine the equation of the graph, *f*: $y = -x - 3$

2.2 **Use the graph** to answer the following questions:

2.2.1 Find the value of *y* if $x = -4$ $y = 1$ see A

2.2.3 For what value of *x* is $y = -2$ $x = -1$ see B

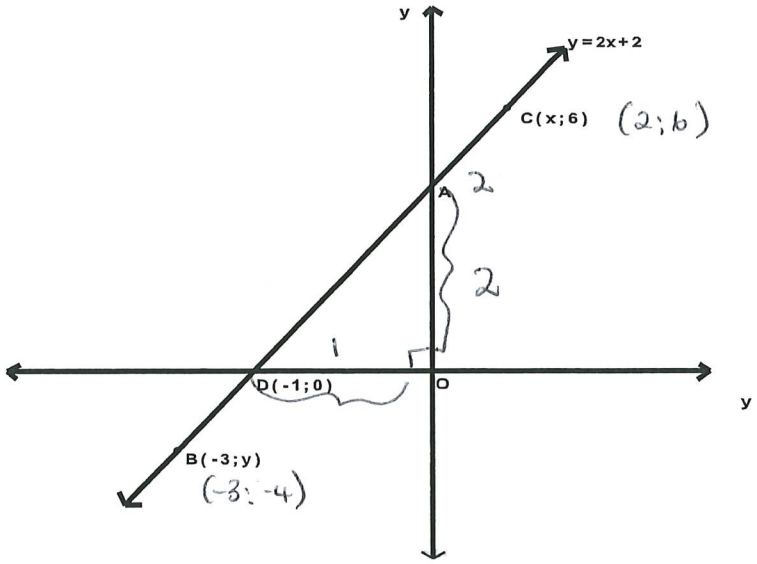
2.3 On the same system of axes, draw the graph of $y = 3x$

Use your graph to answer the following questions:

2.3.1 Find the value of *y* if $x = 1$ $y = 3$ see C

2.3.2 For what value of *x* is $y = 4$ $x = \frac{4}{3}$ see D

3. On the set of axes below, the graph of $y = 2x + 2$ has been drawn.



3.1 Write down the coordinates of A

(0;2)

3.2 Determine the *y* coordinate of B.

$y = 2(-3) + 2$
$= -4$

3.3 Determine the *x* coordinate of C

$6 = 2x + 2$
$4 = 2x$
$2 = x$

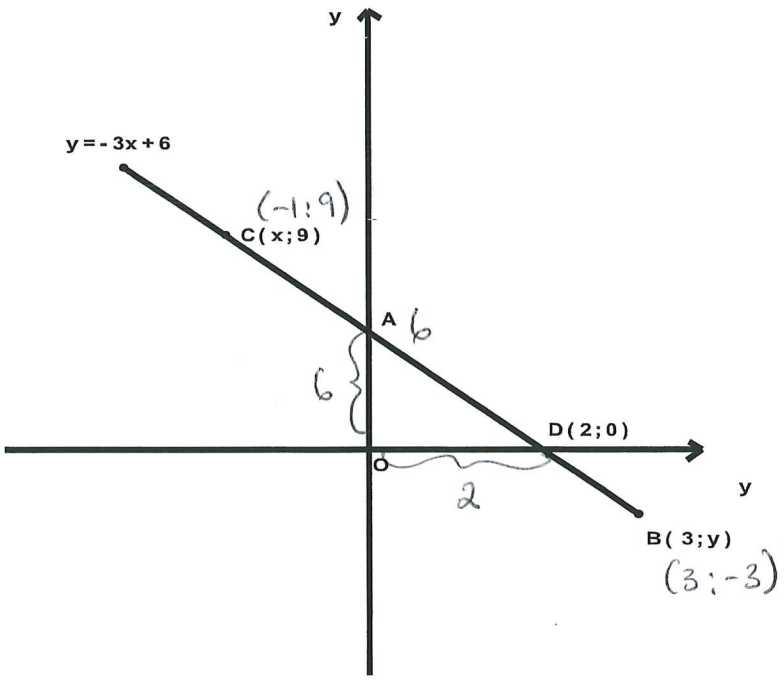
C (2;6)

3.4 Determine the length of AD

$2^2 + 1^2 = AD^2$ (pythag)
$5 = AD^2$
$AD = \sqrt{5}$ units

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4. On the set of axes below, the graph of $y = -3x + 6$ has been drawn.



4.1. Write down the coordinates of A

A (0;6)

4.2 Determine the *y* coordinate of B.

$y = -3(3) + 6$
$= -9 + 6 = -3$

4.3 Determine the *x* coordinate of C

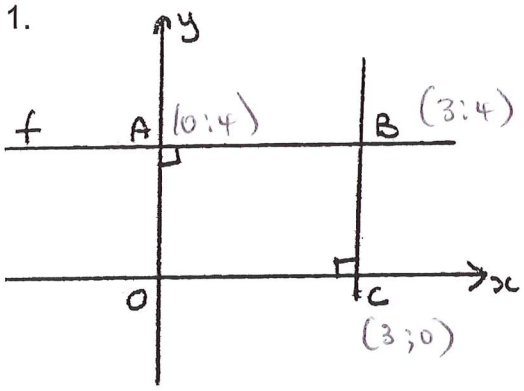
$9 = -3x + 6$
$3 = -3x$
$-1 = x$

4.4 Determine the length of AD

$AD = \sqrt{6^2 + 2^2}$ (Pythag)
$= \sqrt{40}$
$= 2\sqrt{10}$ units

EXERCISE B

1.



Equation of AB is $y = 4$

Equation of BC is $x = 3$

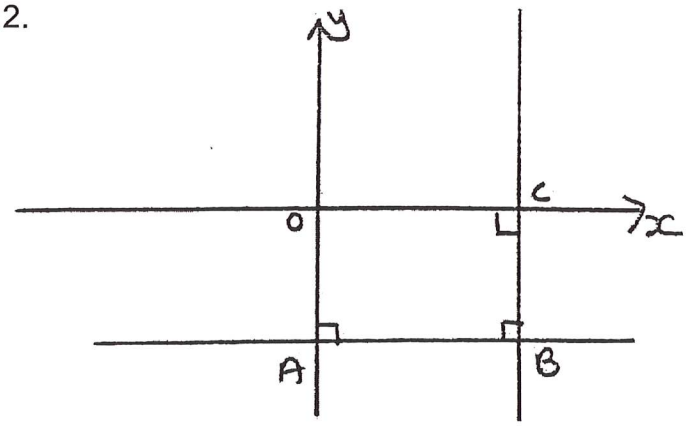
a. the length of AO 4

b. the length of OC 3

c. the co-ordinates of B (3;4)

d. the length of OB. 5 (pythag)

2.



Equation of AB is $y = -9$

Equation of BC is $x = 12$

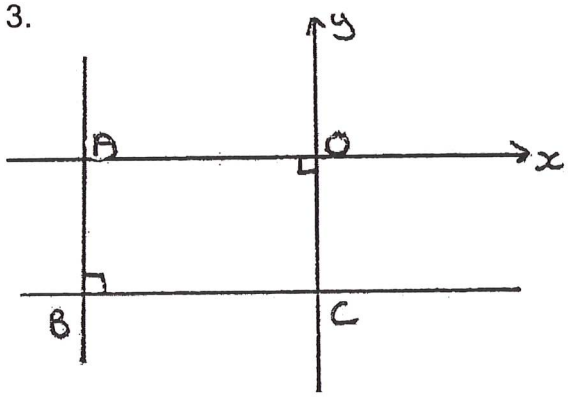
a. the length of AO

b. the length of OC

c. the co-ordinates of B

d. the length of OB.

3.



B is $(-5 ; -12)$

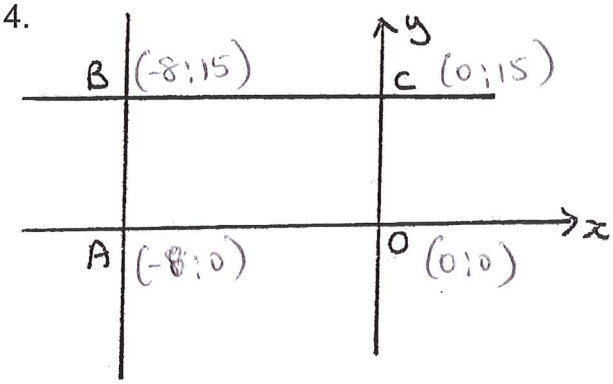
Find:

a. Equation of AB

b. Equation of BC

c. the length of BO.

4.



B is $(-8 ; 15)$

Find:

a. Equation of AB $x = -8$

b. Equation of BC $y = 15$

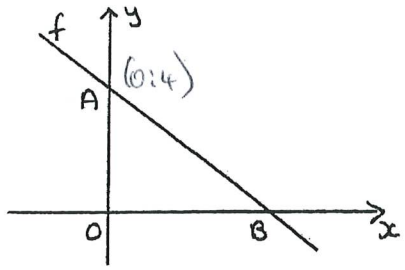
c. the length of BO. $\sqrt{15^2 + 8^2}$ =BO

BO = 17

Remember that distance is always positive.

EXERCISE C

1.



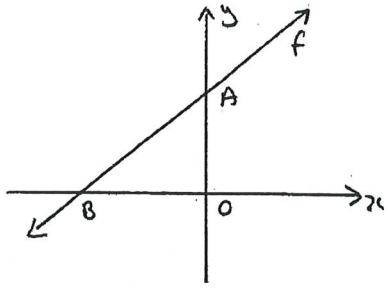
$y = -2x + 4$

Find: a. Length of AO 4

b. Length of OB.

B? $y = 0$ $x = 2$ $\therefore OB = 2$

2.

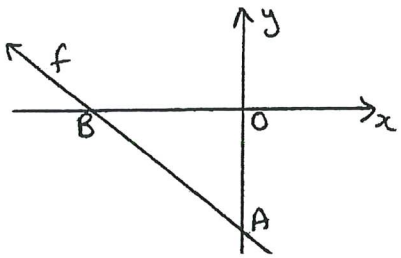


$y = x + 3$

Find: a. length of AO 3

b. length of OB. 3

3.



Given: $y = -x - 2$

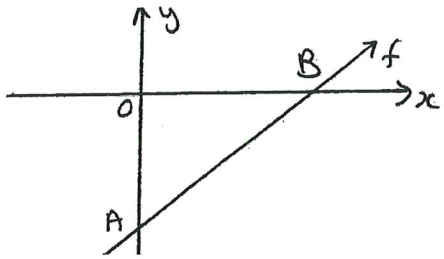
Find: a. length of OA 2

b. length of OB 2

c. length of AB $\sqrt{8}$

(Leave answer in surd form.)

4.



Given: $y = \frac{1}{2}x - 2$

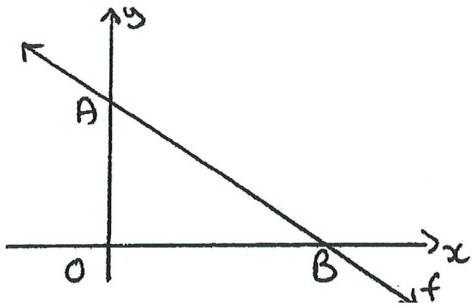
Find: a. length of OA 2

b. length of OB 4

c. length of AB $\sqrt{20}$

(Leave answer in surd form.)

5.



Given: $y = -\frac{1}{5}x + 1$

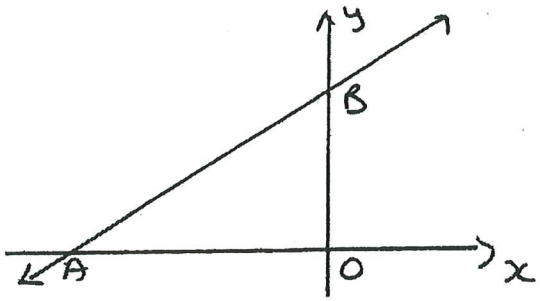
Find: a. length of OA 1

b. length of OB 5

c. length of AB $\sqrt{26}$

(Leave answer in surd form.)

6.



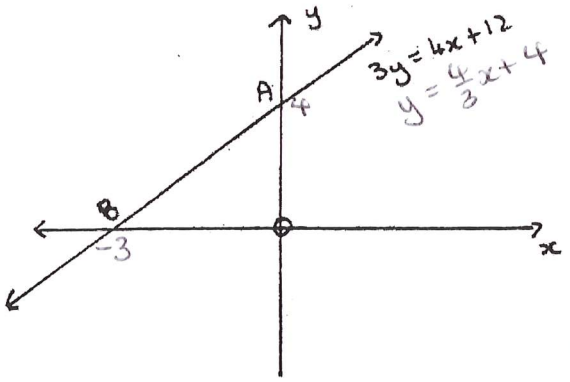
Given: $y = \frac{2}{3}x + 2$

Find: a. length of OB 2

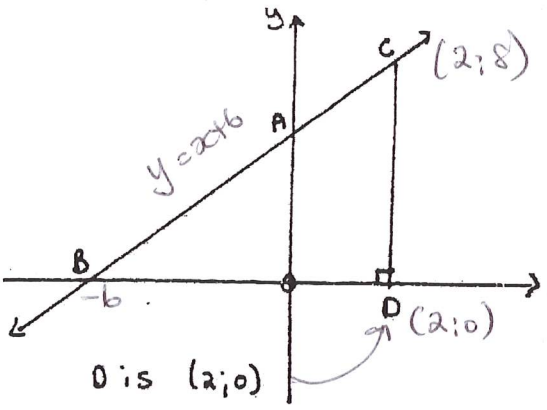
b. length of OA 3

c. length of AB $\sqrt{13}$

(Leave answer in surd form.)



y=x+ 6



Find:

a. OA 4

b. OB 3

c. AB 5 (pythag)

Find:

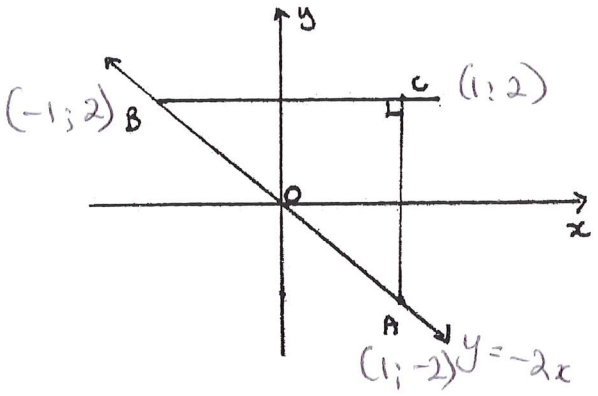
a. CD 8

b. BD 8

c. OA 6

d. BC $\sqrt{8^2+8^2}$ $= \sqrt{128}$ $= 8\sqrt{2}$

3.



The equation of AB is $y = -2x$

and C is $(1 ; 2)$.

Find: (Leave answer in surd form.)

a. AC ~~DE~~ 4 units.

b. BC 2 units

c. AB $\sqrt{20}$ $= 2\sqrt{5}$